

OJAS TASKAR

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EDUCATION

JOHNS HOPKINS UNIVERSITY | Baltimore, Maryland, United States

Master of Science, Robotics | Expected 05/2026 | GPA: 3.8/4.0

- Coursework: Computer Vision, Algorithms for Sensor-Based Robotics, Machine Perception, Vision as Bayesian Inference, Medical Image Analysis, Deep Learning for Cognitive Neuroscience, Trajectory Generation for Space Systems
- Teaching Experience: **GTA:** Algorithms for Sensor Based Robotics (Fall 2025), **GTA:** AI Essentials for Business (Spring 2025)

VEERMATA JIJABAI TECHNOLOGICAL INSTITUTE | Mumbai, India

Bachelor of Technology, ECE | 06/2024 | CGPA: 8.78/10

Bachelor of Technology in **Electronics and Telecommunication** with *Distinction*

TECHNICAL SKILLS

- Languages: Python, C++, Julia, JAX, MATLAB
- Packages: PyTorch, PyTorch Lightning, Tensorflow, CUDA, TensorRT, OpenCV, CMake, GTSAM, ROS2, MoveIt, Gazebo
- Softwares: Hugging Face Diffusers, MuJoCo, IsaacSim, AWS, Wandb

RESEARCH EXPERIENCE

Research Intern | Image Analysis & Communications Laboratory (IACL), JHU | Prof. Jerry Prince | June 2025 - August 2025

- Built a bidirectional diffusion model to synthesize **paramagnetic rim lesions** on healthy brain susceptibility-weighted images.
- Modified a SOTA **multi-view DDIM inversion & fusion** pipeline conditioned on lesion masks for **10x faster synthesis output**.
- Presented review of 20+ automated rim lesion detection papers to lab scholars, highlighting key literature gaps.

Graduate Research Assistant | BIGSS Lab, JHU | Dr. Amir Kheradmand | September 2024 - June 2025

- Developed multi-camera 6DOF pose estimation for sub-millimeter tracking of patient's head, for robot-assisted TMS.
- Built a Bayesian plane fitting algorithm to solve monocular image depth ambiguity, reducing depth error from **40mm** to **6mm**.
- Deployed **deformable Gaussian Mixture Models**, writing custom PyTorch plugins for point cloud to MRI registration.
- Created a dynamic scene graph with pose graph optimization for device-agnostic surgical instrument tracking using Microsoft HoloLens2. Co-authored a **manuscript** submitted to **IEEE VR 2026**.

Research Intern | KIREAP, Inc. India | Dr. Omkar Halbe | January 2024 - June 2024

- Prototyped algorithm development of a drone vision-based landing system, reducing AprilTag detection **false positives** by **20%**.
- Utilized AprilTag pattern arrangement and lightweight CNNs to design passive pad tracking system, with **77% trial success**.
- Benchmarked instance segmentation pipelines on Jetson Nano, writing custom CUDA kernels for **35% throughput increase**.

PROJECTS

Neuro-image Registration and Synthesis | Medical Image Analysis Course Project | Prof. Jerry Prince | Spring 2025

- Developed a pipeline for **automated diffeomorphic registration** between T1w/T2w and FA/ADC diffusion tensor images using ANTsPy, improving robustness to imaging artifacts and voxel resolution mismatch.
- Tuned a pre-trained **SegResNet 3D convolution** network on registered image pairs, achieving ventricular area DICE of **0.92**.
- Evaluated registration quality, leveraging **Jacobian of Deformation Field**, **Inverse Consistency Error** and intra-region **DICE loss**.

Cardiac Ejection Fraction Estimation | Medical Image Analysis Course Project | Prof. Jerry Prince | Spring 2025

- Trained a modified- UNet for left ventricular (LV) segmentation from echocardiogram videos of over 7,460 patient recordings.
- Engineered an empirical area-to-volume relationship to accurately estimate ejection fraction (EF), attaining an RMSE of **1.42%**.
- Formulated a novel PCA-based algorithm to detect **ventricular axis** in 2D echocardiogram slices from video sequences.

Contrastive Pre-Training for Instance Classification | Computer Vision Course Project | Prof. Kapil Katyal | Fall 2024

- Wrote a SimCLR-based pretraining pipeline with **γ -aware & InfoNCE loss** for anomaly detection, with **42%** top-5 validation loss.
- Developed downstream vision transformers (ViT) to output class labels, **improving precision** over VGG19 or ResNet.
- Performed comparative analyses of data augmentation techniques and subsequent impact on validation loss, determining gaussian blur and random image crop having **78% correlation** to positive detection.

Bachelor's Thesis | Swarm Area Search Using Chaos-Based Secured Communication | Dr. Harish Chaudhari, VJTI | Summer 2023

- Designed a C++ based central system for heterogenous drone swarm, with dynamic Voronoi cell generation for search-radius aware planning, attaining **57% trial accuracy**.
- Deployed a TSP with HK optimization solution, leading to 5% less collision cases and reduced area search overlap.
- Built a chaotic keyed communication architecture on MATLAB and Simulink, decreasing efficiency of intrusion attacks by **30%**.